Online word prediction and offline probability judgment processes are partially shared

Yutong Zhang & E. Matthew Husband (yutong.zhang, matthew.husband@ling-phil.ox.ac.uk)
Faculty of Linguistics, Philology and Phonetics, University of Oxford

Introduction. The idea that comprehenders compute the probability for upcoming words has gained much support over the last twenty years. Most prominently, Surprisal Theory proposes that the reading time (RT) of a word is proportional to its surprisal, the negative log probability of the word given the current context (Levy, 2008). How comprehenders arrive at these underlying word probabilities is not well understood. Contextual features that affect word probabilities are often simply operationalized and measured by LLMs/cloze norming, and the processes used by LLMs and in cloze tasks to generate the probabilities that underlie surprisal remain obscure.

Outside of psycholinguistics, judgment and decision-making (J-DM) research finds that context affects subjective probability judgment (PJ) and has investigated factors affecting how humans arrive at these probabilities. Overt mention of a category exemplar, for example, systematically biases PJ: Typical exemplars lead to higher PJs and atypical exemplars lead to lower PJs (Sloman et al., 2004; Tversky & Koehler, 1994). The typical exemplar effect also appears to be more fragile in superordinate-level target categories (*instrument*) compared to basic-level ones (*snake*).

If judgment making and sentence processing both rely on probability inference, are the processes underlying offline probability judgments shared with online word predictions? To investigate this, we manipulated unpacking and exemplar typicality in a probability judgment and a self-paced reading task. We predicted that by-item PJ changes correlate with by-item RT changes ($\Delta PJ \propto -\Delta RT$, Table 1) if probability judgment shares processes with word prediction.

Method. An offline probability judgment task (N=36) and an online self-paced reading task (N=60) were conducted on the same 60 experimental items, closely modelled on traditional J-DM manipulations (using Peelle et al., 2020, cloze corpus; 30 basic-level (*snake*) + 30 superordinate-level (*instrument*) target word items). We manipulated exemplar typicality in Typical (*cobra/piano*) and Atypical (*cottonmouth/banjo*) Unpacked conditions, compared to a Packed baseline condition with no exemplar mentioned (see Figure 1). Each item was presented as a probability judgment question in the judgment task (*What is the probability that…*), and as a sentence in the reading task. Participants provided PJs on a scale of 0-100%, and we measured RTs word by word.

Results. While we focused on the by-item relationship between PJs and RTs, briefly, we found standard typicality effects on PJs, including the fragility of typical exemplar effects in superordinate vs. basic-level categories, and general slowdowns in RTs for unpacked contexts regardless of typicality, especially for basic-level target words (Figure 1). Across items, changes in PJs and RTs from Atypical exemplars showed significant correlation in both basic-level (β=-1.24, p=.039) and superordinate-level (β=-1.41, p=.012), with no significant correlation from Typical exemplars (basic-level: β=-0.10, p=.858; superordinate: β=-0.06, p=.910) (Figure 2).

Discussion. Probability judgment results replicated unpacking biases reported in the J-DM literature. More importantly, changes in offline PJs correlated with changes in RTs, but only when categories were unpacked by atypical exemplars. Given that typical exemplar effects on PJs are more fragile, it may be that they are more related to higher-level reasoning processes, while atypical exemplar effects on PJs are driven more by basic cognitive processes that are also deployed in comprehension that determine upcoming word probability. This pattern suggests that the processes underlying online word prediction and offline probability judgments may be partially shared, with important differences that require further exploration.

Table 1: Predictions for unpacking biases in probability judgment and word prediction.

	Probability Judgment	Word Prediction
Typical Unpacked	$P(T \mid C) < P(T \mid C, A_{typical})$	$-\log P(W \mid C) > -\log P(W \mid C, A_{typical})$
Atypical Unpacked	$P(T \mid C) > P(T \mid C, A_{atypical})$	$-\log P(W \mid C) < -\log P(W \mid C, A_{atypical})$
Δ = Unpacked - Packed	$\Delta PJ = P(T \mid C, A) - P(T \mid C)$	$\Delta RT \propto (-\log P(W \mid C, A)) - (-\log P(W \mid C))$

Figure 1: Average probability judgments and average reading times by word.

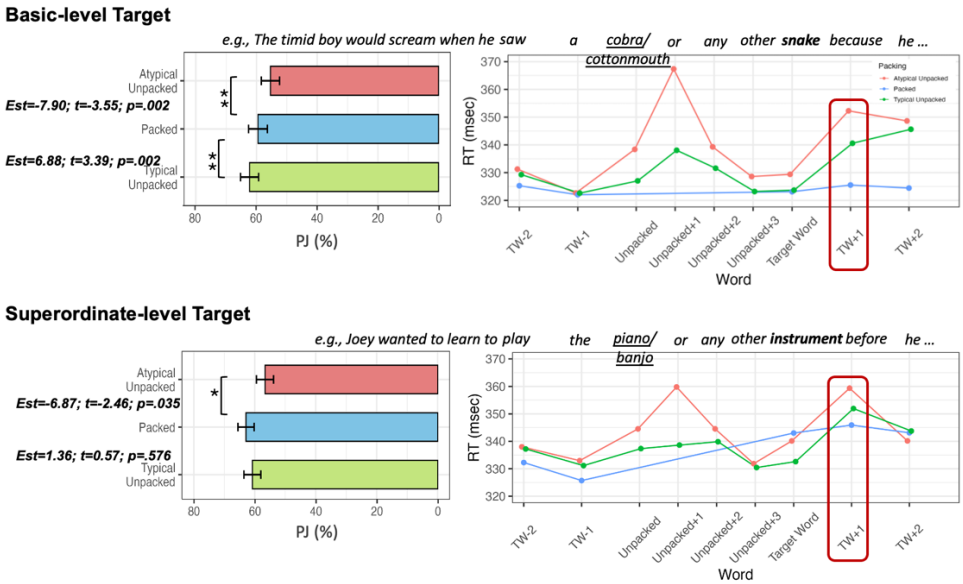
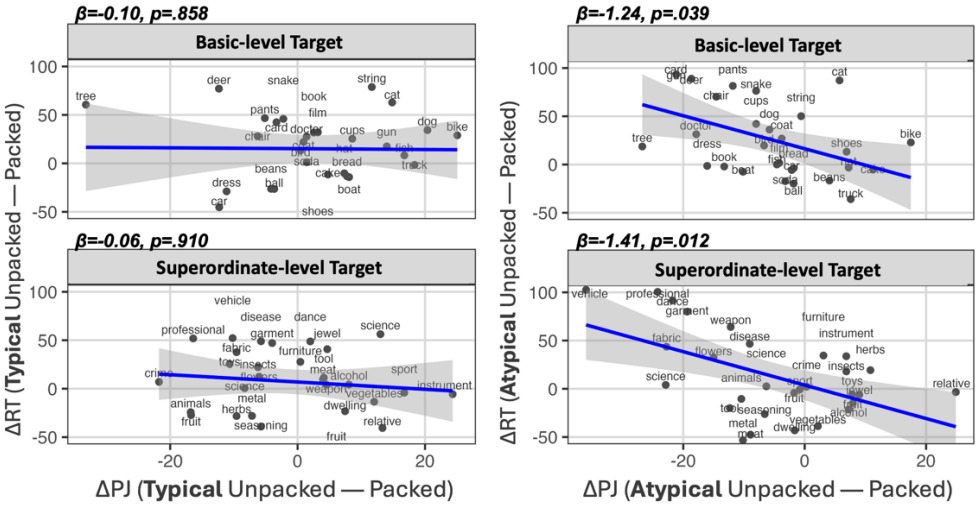


Figure 2: Item-level interaction between ΔPJ and TW+1 ΔRT by typicality and category level.



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